

mlxburn – Firmware Image Generator and Burner

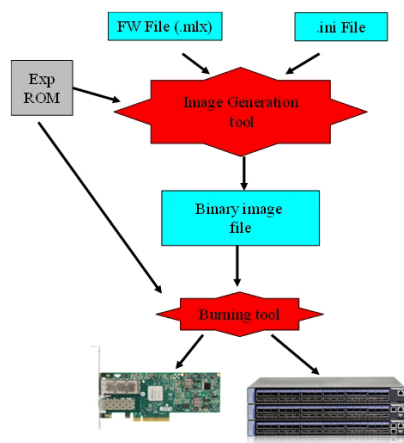
mlxburn is a tool for firmware (FW) image generation and/or for burning a firmware image to the Flash/EEPROM attached to an NVIDIA device. Both functions or a single function of mlxburn can be activated by means of command line options (see [mlxburn Synopsis](#)). It can also query for firmware attributes (e.g., firmware version, GUIDs, etc.) and VPD info of adapter cards and switch systems.

mlxburn allows for customization of standard NVIDIA firmware for OEM specific needs (e.g., a specific adapter board type). See [Customizing Firmware](#).

Generating and Burning Firmware

The mlxburn firmware update flow is composed of two separate stages: image generation and image burning. In the image generation stage, a given NVIDIA firmware release in .mlx format is processed together with a board-specific configuration (.ini) file to generate a 'burnable' firmware image. This image is burnt to the Flash/EEPROM attached to an NVIDIA device in the second stage. The burning process retains device specific data such as GUIDs, UUIDs, MACs, VSD, and BSN. Also, the burn process is failsafe by default.

FW Generation and Burning



mlxburn runs both stages by default, but it may perform only one by means of command options. If '-wrimage' is specified (see [mlxburn Synopsis](#)), only image generation is performed. Specifying the '-image' option skips the image generation stage and loads the provided image (generated in a previous run of mlxburn using the '-wrimage' option).

Customizing Firmware

An NVIDIA firmware image can be customized (usually) to fit a specific board type. The customization is done by using a FW parameter-set file in the image generation stage. This file has a .ini format. Each parameter-set file has a unique parameter-set ID (PSID) which

During a device FW update, mlxburn reads the PSID from the device and uses the corresponding .ini file when generating the FW image. mlxburn searches for the files in the same directory of the FW release. When mlxburn is used to generate an image file, or when no corresponding parameter-set file is found, the user should explicitly specify which parameter-set file to use.

To produce an image file the user needs to provide the option ‘-wrimage <target file>’. To actually burn the image to the Flash/EEPROM attached to an NVIDIA adapter or switch device, the user needs to specify the option ‘-dev <mst device>’ (see the synopsis section below).

If run in burning mode, mlxburn auto-detects the firmware parameter-set with which the device was previously burnt. It locates and uses this parameter-set file to generate the appropriate image for the device (by merging the FW release with the specific parameter-set required).

To inhibit image generation, the ‘-image <pre-generated-image-file>’ should be used. It instructs mlxburn to use the given file for burning the device.

mlxburn Synopsis

```
1  #mlxburn [-h][-v] <-dev mst-device|-wrimage fw-image> <-fw mellanox-fw-file|
2  [-dev_type device type] [-exp_rom <exp_rom_file>][-exp_rom_dir <exp_rom_dir>
3  [-vpd_set_keyword <keyword-assignment>] [-set_pxe_en <(port1|port2)=(enable|
```

Note: The “-fwver” flag is not supported in Connect-IB, Switch-IB, ConnectX-4, and ConnectX-5 devices.

where:

-dev_type <mellanox-device-number>	mlxburn must l device type in c work properly. flag if device ty detection fails. Example: -dev_ Supported NVI device types: • HCAs/NI 25418, 2 25448, 2 26478, 2 26438, 2 4103, 41 4117, 41 41680, 4

	- Switches 48437, 4 52000, 5 53000, 5
-fw <mellanox-fw-file>	Specify NVIDIA Firmware File to extension is .m
-conf <parameter-set-file>	FW configurati Needed when g image (not usin or if configurat detection fails.
-conf_dir <dir>	When specified detected confi will be looked f given directory the firmware fi Applicable for l operation.
-gb_bin_file <gb_bin_file>	Integrate the g binary file to th
-dev <mst-dev>	Burn the image given mst devic
-exp_rom <exp-rom-file>	Integrate the g expansion rom FW image. The may be in .img (raw binary) for If the exp set to "A expansio auto dete the files the exp_ (see belo NOTE: Exp rom detection is do devices that ar

	burned with an rom image. If "-exp_r" is specified with no expansion file would be used. If no expansion file is specified, no expansion file is used. To add expansion files manually, supply the expansion file to use.
-exp_rom_dir <exp_rom_dir>	The directory in which to look for expansion files when "-exp_r" is specified. By default, expansion files are searched for in the directory <fw file directory>/exp.
-force	None interactive. Assume "yes" for all questions.
-format <BINARY IMAGE>	Specify which image format to use. Can be used with the -wrim option. Default is BINARY.
-fw_dir <dir>	When specified, the directory detected fw file will be looked for in the directory. Applicable for the burn operation.
-conf_dir_list <dir1,dir2,...,dirn>	When specified, the directory detected configuration files will be looked for in the given directories in the firmware directory. Applicable for the burn operation.
-fwver	When a device is connected, display current firmware version.

	When a FW file (fw flag): Display the FW version. Note: The “-fw” flag is not supported on IB devices.
-h	Display a short help message.
-image <fw-image-file>	Do not generate the image; use the given fw image.
-img_dir <image directory>	Do not generate the image; select the image from the *.bin images in the given directory.
-nofs	When specified, the file system creation process will not be performed.
-nofs_img	When specified, the image will not be formatted and burn process will be failsafe.
-striped_image	When specified, the image will be in striped format, and will be in striped format when queried.
-query	Query the HCA device for firmware version, e.g. Firmware Version, GUIDs, etc. In addition to the above flags, Mlxburn also accept the following flags/options, which are passed to the underlying burning tool: -banks -use_bank -skip_is -skip_is -mac(s) -guid -sysguid -vsd -ndescribe -pe_i2c -se_i2c

	-is3_i2c -no -uid(s -log -blan flash_params -allow_psid_ch no_flash_verify -use_image_ro override_cache -use_image_gu See the flint to documentation gen switches/E burning option
-v	Print version in
-V <INFORM WARNING DEBUG>	Set verbosity le WARNING
-vpd ^{1,2}	Display the rea section of the l (Vital Product I given device
-vpd_rw _{1,2}	(on Linux only) the read/write PCI VPD of the
-wrimage <fw-image-file>	Write the imag given file.
-s, --i2c-secondary <address>	Change the I2C secondary add

Note 1. The VPD query may not be enabled on certain board types. Also, VPD operations are available only for devices with a PCI interface.

Note 2. Running multiple VPD access commands in parallel on the same device, by mlxburn or any other VPD access tool, may cause the commands to fail. VPD access commands should be run one at a time.

Connect-IB, Switch-IB, Switch-IB 2, NVIDIA Spectrum, ConnectX-4, and ConnectX-4 Lx Initial Burning Options

The following options are relevant when generating an image for initial burning. The image contains the VPD and the GUIDs that are in a read-only area on flash.

```
1 [ -vpd_r_file <vpd_r_file>] [ -base_guid <GUID>]
```

where:



-vpd_r_file <vpd_r_file>	Embed the given VPD Read-Only section in the generated image. The vpd_r_file should contain the vpd read only section and the first dword of the vpd writeable section. The file is in binary format, and its size must be a multiple of 4 bytes. Please refer to PCI base spec for VPD structure info.
-base_guid <GUID>	Set the given GUID as the image base GUID. The base GUID is used to derive GUIDs and MACs for the HCA ports. It is assumed that 16 GUIDs (base_guid to base_guid +

	15) are reserved for the card. Note: On ConnectX-4/ConnectX-4 Lx/ConnectX-5/ConnectX-5 Ex, only GUIDs will be derived according to the HCA configuration.
-base_mac <MAC>	Set the given MAC as the image base MAC. The base MAC is used to derive MACs for the HCA ports according to the device configuration (ConnectX-4/ConnectX-4 Lx/ConnectX-5/ConnectX-5 Ex).
-vsd <string>	Write this string, of up to 208 characters, to VSD section.

Additional mlxburn Options

The following is a list of additional options. Please see [.mlxfwmanager – Firmware Update and Query Tool v4.18.1](#) for the HCA options.

```
-banks -use_image_ps -skip_is -mac(s) -guid(s) -sysguid -ndesc -bsn -
```

```
-override_cache_replacement -ocr -ignore_dev_data -use_dev_rom -no_fw_ct
```

The arguments of the `-quids` and `-macs` flags must be provided within quotation marks; for example, `mlxburn -macs "0002c900001 0002c900002"`.

Examples of mlxburn Usage

Host Channel Adapter Examples

To update firmware on an MT25408 ConnectX adapter device with the configuration file (.ini) auto-detected, enter:

```
1 # mlxburn -fw ./fw-ConnectX3-rel.mlx -dev /dev/mst/mt4099_pci_cr0
```

To generate a failsafe image file for the same adapter above without burning, enter:

```
1 # mlxburn -fw ./fw-ConnectX3-rel.mlx -conf ./MCX354A-FCB_A2-A5.ini -wrimage
```

To update firmware on the same adapter above with the configuration file (.ini) explicitly specified, enter:

```
1 # mlxburn -fw ./fw-ConnectX3-rel.mlx -dev /dev/mst/mt4099_pci_cr0 -conf ./CX
```

ConnectX-5 Examples

To generate a failsafe image file for ConnectX-5 device without burning, enter:



```
# mlxburn -fw FW/fw-ConnectX-5.mlx -conf FW/CX515A-CCA_Ax.ini -wrimage fw-ConnectX-5-CX515A-CCA_Ax.bin -base_guid 0x002c90330123e00
```

To update firmware on a ConnectX-5 device, enter:

```
1 # mlxburn -i fw-ConnectX-5-CX515A-CCA_Ax.bin -d /dev/mst/mt4115_pciconf0
```

ConnectX-4 Examples

To generate a failsafe image file for ConnectX-4 device without burning, enter:

```
1 # mlxburn -fw FW/fw-ConnectX-4.mlx -conf FW/MCX456A-ECA_Ax.ini -wrimage fw-C
```

To update firmware on a ConnectX-4 device, enter:

```
1 # mlxburn -i fw-ConnectX-4-MCX456A-ECA_Ax.bin -d /dev/mst/mt4113_pciconf0
```

ConnectX-4 Lx Examples

To generate a failsafe image file for ConnectX-4 Lx device without burning, enter:

```
1 # mlxburn -fw FW/fw-ConnectX4Lx.mlx -conf FW/MCX4131A-GCA_Ax.ini -wrimage fw
```

To update firmware on a ConnectX-4 Lx device, enter:

```
1 # mlxburn -i fw-ConnectX-4LX-MCX4131A-GCA_Ax.bim -d /dev/mst/mt4117_pciconf0
```

Connect-IB Examples

To generate a failsafe image file for Connect-IB device without burning, enter:



```
# mlxburn -fw FW/fw-ConnectIB.mlx -c FW/MCB194A-FCA_A1.ini -wrimage fw-ConnectIB-  
MCB194A- FCA_A1.bin -base_guid 0x0002c903002ef500
```

To update firmware on a Connect-IB device, enter:

```
1 # mlxburn -i fw-ConnectIB-MCB194A-FCA_A1.bin -d /dev/mst/mt4113_pciconf0
```

SwitchX Switch Examples

Burn an MT51000 switch system using the In-Band access method:

```
1 # mlxburn -dev /dev/mst/SW_MT51000_000002c900002100_lid-0x000E -fw ./fw-sx.m
```

Generate an MT51000 image and perform an In-Band update of the device with LID 0xE:

```
1 # mlxburn -dev lid-0x000E -fw ./fw-sx.mlx
```

Generate and burn a new MT51000 via I2C:
Set the I2C network to access the SwitchX switch.

```
1 # mlx2c -d /dev/mst/mtusb-1 p SX
```

Burn the new image (the flash is still blank) specifying the Node GUID, system GUID, base MAC and Switch MAC. Note that 4 guids (in quotes) should be specified as an argument to the -guids flag. The 2 middle GUIDs are ignored by SwitchX and should be set to 0.

```
1 # mlxburn -d /dev/mst/mtusb-1 -fw ./fw-sx.mlx -conf MSX6025F_A1.ini -guids "
```

NVIDIA Spectrum Examples

To generate a failsafe image file for a NVIDIA Spectrum device without burning, enter:



```
mlxburn -fw FW/fw-SwitchEN.mlx -c FW/MSN2700-Cxxx_Ax.ini
-wrimage fw-Spectrum-MSN2700-Cxxx-_Ax.bin -base_guid
e41d2d030045a240 -base_mac 0000e41d2d45a240
```

To update firmware on a Spectrum device, enter:

```
1 mlxburn -i fw-Spectrum-MSN2700-Cxxx_Ax.bin -d /dev/mst/mt52100_pciconf0
```

Switch-IB Examples

To generate a failsafe image file for a Switch-IB device without burning, enter:

```
1 mlxburn -fw FW/fw-SwitchIB.mlx -c FW/MSB7700-E_Ax.ini -wrimage fw-SwitchIB-M
```

To update firmware on a Switch-IB device, enter:

```
1 mlxburn -i fw-SwitchIB-MSB7700-E_Ax.bin -d /dev/mst/SW_MT52000_00001111101a
```

Switch-IB 2 Examples

To generate a failsafe image file for a Switch-IB 2 device without burning, enter:

```
1 mlxburn -fw FW/fw-SwitchIB-2.mlx -c FW/MSB7800-Exxx_Ax.ini -wrimage fw-Switc
```

To update firmware on a Switch-IB 2 device, enter:

```
1  mlxburn -i fw-SwitchIB-2- MSB7800-Exxx_Ax.bin -d /dev/mst/mt53000_pciconf0
```

Exit Return Values

The following exit values are returned:

- 0 - successful completion
- >0 - an error occurred

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